

Sound setup: what we get for the money

I would start with about \$700 for selected tones below 20 kHz. That buys a speaker, measurement microphone, USB interface and our own sound-level reference. If we need every SONIC tone through 32 kHz, the wider-band setup below is about \$11,000–\$12,000, pending a TDT quote. We can use our computer and lab neural recorder in either case.

Our options

Option	Working budget	What it gets us
PC + USB audio	About \$450 with a borrowed reference; \$700 with our own	The simplest candidate for selected 1.4–20 kHz tones. Speaker and microphone specifications stop short of SONIC’s 32 kHz upper end.
Arduino	\$100–\$150 demo; about \$850 with the better audio gear	Board-controlled playback and trigger signals. The small demo speaker/mic are uncalibrated; the \$850 version keeps the equipment from the audible setup. Firmware and testing required.
FPGA	About \$1,200 with audible gear; boards alone \$134.38	Custom playback, capture and digital markers. Same speaker/mic limit as the audible setup. Requires FPGA code, clock setup and connection design.
Full SONIC range	About \$11,000–\$12,000; quote dependent	A speaker and measurement chain covering 1.4–32 kHz, with NI hardware timing. A closer engineering starting point to the paper.

What precision can we actually claim?

Tone frequency. Cheap audio hardware can generate the chosen tones, but I found no guaranteed clock error in Hz for the proposed interface. NI specifies a 50 ppm clock: that contributes about **±0.5 Hz at 10 kHz or ±1.6 Hz at 32 kHz**, provided we generate the waveform using the actual sample rate returned by the driver. These are clock calculations, not measured sound accuracy. [NI specification](#); [actual sample rate](#).

Sound level at the ear. The [reference calibrator](#) is specified at **±0.5 dB at 1 kHz**. It does not guarantee every tone at the ear is within ±0.5 dB. We still use the microphone’s correction file and measure each tone at the chosen position. Whole-system SPL accuracy is unmeasured for every option.

Sound-to-brain timing. At 192,000 samples/s, waveform samples are **5.2 microseconds apart**; a proposed 100,000-sample/s Arduino design has a 10-microsecond grid. Neither is a claim of neural alignment accuracy. Record physical markers and check sound arrival/clock drift with our recorder. Faster digital triggers do not remove speaker or acoustic delay.

What SONIC tells us

[SONIC v1](#), [Methods 6.1–6.2](#), [p.11](#): USB-6361 → SA1 → MF1; 1.4–32 kHz initial tones, 10 ms benchmark tones, 2 ms ramps, nominal 192,000 samples/s and about 80 dB at the ears. It used PTP/10 MHz synchronization and recorded output/trial references. It does **not** report measured ±Hz, ±dB or sound-to-neural jitter. Its sheep were awake; our sedated-sheep responses remain unverified. The large rack in the slides is not needed for this USB approach.

The affordable setup and DIY parts

Blue part names are purchase links. Prices below were checked on October 9. Item prices exclude checkout tax/shipping; the rounded working budgets include estimated reserves. No new computer or neural recorder is counted.

The approximately \$700 audible setup

Buy one unless stated	USD	Seller / what it does
Behringer UMC204HD	97.98	eBay, Sweetwater; new. Playback, microphone preamp and recording.
JBL 305P MkII, single speaker	122.39	Reverb; new/open box, power cord included. Amplifier built in.
Dayton EMM-6 microphone	66.93	eBay, Parts Express; new. Individual frequency correction file.
Hosa HMIC-010 XLR cable	17.99	Sweetwater; microphone cable.
Two Hosa HSS-005 TRS cables	29.90	\$14.95 each; speaker and electrical playback monitor.
Gator RI-MICTP-FBM stand	34.99	Sweetwater; holds the microphone position.
REED R8090 calibrator	219.00	DigiKey / REED; new, in stock. Borrow instead if available.
Parts subtotal	589.18	\$370.18 if we borrow the calibrator.

$\$589.18 + \$100 \text{ shipping/tax reserve} = \textbf{\$689.18, rounded to \$700}$. With a borrowed reference: $\$370.18 + \$80 \text{ reserve} = \textbf{\$450.18}$. Reuse a stable speaker surface and the interface's supplied USB cable. The recorder-specific timing cable/adaptor is still unselected and additional.

The [speaker](#) is specified through 20 kHz; the [mic](#) through 20 kHz. Its half-inch tip nominally fits the calibrator; check the seal. The basic calibrator does not include a separate traceable certificate. Disable microphone monitoring, automatic gain and sound enhancement.

Can we build the playback and recording ourselves? Yes.

For a first **bench prototype**, use these four parts:

Part	USD	Job
Arduino GIGA R1 WiFi	69.60	Timed playback, capture and markers.
Adafruit MAX4466 mic/preamp	6.95	Basic microphone signal for the board's input.
Adafruit PAM8302 amplifier	3.95	Drives the small speaker.
Adafruit 3351 speaker	3.95	Small 4-ohm speaker; usable tone band unverified.

\$84.45 in parts, plus \$20–\$40 wiring/power allowance, then shipping/tax: about **\$100–\$150**. This tests the program and timing idea; it is not a calibrated replacement for the shopping list above. The board's 12-bit DAC/up-to-16-bit ADC do not establish effective analog accuracy.

Arduino with the better audio gear: keep the \$589.18 setup for microphone power/capture, add the GIGA at \$69.60, \$75 for output coupling/filtering and marker parts, and \$100 reserve: **\$833.78**. Connection parts require design.

FPGA version: replace the \$97.98 USB interface with a [Cmod A7](#) (\$104), [I2S2 audio board](#) (\$30.38) and [MM-1 mic preamp](#) (\$325). With the same speaker/mic/reference: \$950.58 hardware + \$100 connection allowance + \$150 reserve = **\$1,200.58**. Requires adapted cables, FPGA code and testing; labor extra.

If we need the full SONIC sound range

Computer → NI USB output → amplifier → speaker.

Sound at ear → microphone → preamp → NI input.

Our lab recorder still records the brain, with a compatible physical timing reference.

Part, one each	USD	Buying status
NI USB-6361, 781442-01	3,030.42	Newark; new screw-terminal model, Add to Cart.
Earthworks M50 microphone	1,299.00	eBay; new. B&H alternative . Half-inch calibrator adapter included.
Sound Devices MM-1 preamp	325.00	eBay; used, no returns. Or \$945 new at B&H .
REED R8090 calibrator	219.00	Same reference as page 2; count once.
Priced instruments	4,873.42	Then add the actual TDT quote.
TDT MF2-M speaker kit	1,500 est.	Quote / order . No reliable live used price found.
TDT SA1 amplifier	2,000 est.	Quote / order .
TDT ZB1PS power chassis	750 est.	Quote / order . Powers SA1.

Budget: \$4,873.42 priced instruments + \$4,250 TDT allowance + \$500 wiring/power/mounts allowance = \$9,623.42. With a 15% shipping/tax/uncertainty reserve: **\$11,066.93**. Substituting the new MM-1 gives **\$11,779.93**. These are planning totals, not fixed upper limits or vendor package quotes. A traceable full-band calibration service and labor are additional if required.

MF2 covers 1–80 kHz free field; M50 covers 3 Hz–50 kHz. The MM-1 supplies the required 48 V/10 mA microphone power and has a specified response extending to 50 kHz. Use fixed gain, limiter/high-pass off and correctly wired differential NI input. Two AA batteries can power it. Measure the complete chain; specifications alone do not establish ear-position SPL accuracy.

Where used equipment helps

The \$325 preamp saves \$620 against new. I also found a [USB-6361 at \\$1,395 used](#), but it is a mass-terminated model without supply/accessories: it needs a different cable/breakout. Its price is not a complete replacement cost. A [USB-6211 at \\$600 used](#) is a cheaper engineering candidate, not the paper's device; [settling and shared-input limits](#) need checking at 32 kHz. Substituting it would bring the estimate to **\$8,271.95**, if those checks pass. Neither used NI device is counted in the main total. TDT components remain quote-only.

What our own program would do

Prepare the tone list and waveforms on the computer, then preload a block. Let the Arduino's timer and buffered output schedule the samples; generate the marker from the same hardware event as each tone, rather than a USB message. Capture microphone/output references and save repeated markers with the neural data. [Arduino's library](#) supplies buffered analog I/O, but common ADC/DAC timing and the marker need firmware work. FPGA audio needs similar code plus clock/codec setup; it has no demonstrated precision advantage here.

My choice: start with the \$700 audible setup if those tones answer our question. Build the \$100–\$150 Arduino prototype alongside it if custom control matters. Spend on the full-range speaker/mic when tones above 20 kHz are required. All options need acoustic and recorder timing checks before their final precision can be claimed.